

Modeling Liquidity Spillover Dynamics Across Equity, Cryptocurrency, and Precious Metals Markets Using Econometric Methods

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Abstract—Since the markets are interconnected, a liquidity shock in one asset class could spill over into other asset classes, but no study has measured this across the equity, cryptocurrency, and precious metals markets simultaneously. The literature has been devoted to the study of return and volatility spillovers, and not much is known regarding the transmission of liquidity. The empirical analysis of this paper uses daily data for six assets: S&P 500, NASDAQ, Bitcoin, Ethereum, gold, and silver, spanning the period from January 2020 to December 2024, and computes the Amihud illiquidity ratio. Directional spillovers are measured for all six markets at once by vector autoregression with forecast error variance decomposition. The 60-day rolling correlations detect regime changes over five market periods, while the penalization of illiquid assets in the mean-variance objective function yields the liquidity-adjusted mean-variance portfolio. The FEVD matrix shows that the most important variances are within markets, with significant off-diagonal transmission between equity and precious metals markets. On the worst 5% of equity days, Bitcoin has an average return of -3.3% against gold's near-zero average, which further reinforces gold's weak safe-haven status and denies Bitcoin that classification. The liquidity-adjusted portfolio achieves a Sharpe ratio of 1.15, the same as standard mean-variance, but with the liquidity penalty steering allocation away from high-illiquidity assets. Future research will involve the use of quantile-frequency decomposition to distinguish tail and median liquidity transmission.

Index Terms—liquidity spillover, cross-market connectedness, Amihud ratio, VAR-FEVD, cryptocurrency, precious metals, safe-haven, portfolio optimization, financial contagion

I. INTRODUCTION

During 2020 and beyond, global financial markets underwent transformation much of which was unforeseen in the textbooks. In the wake of the COVID-19 pandemic, the prices of what are generally believed to be quite distinct asset classes all fell together. In March 2020, gold, which is also considered a crisis hedge, sold off slightly along with equities when institutions were forced to sell assets to meet margin calls [10]. Some analysts had dubbed Bitcoin as “digital gold”, but in just days its value had plummeted by more than 50% [8]. Silver traded in line with equity indices rather than gold. What were typically diversification gains were null and void.

Not only return co-movement, but something deeper, as this episode revealed: liquidity dried up simultaneously across multiple markets. As investors began to sell, buyers shied away across the board. This phenomenon is the one that this paper explores: cross-market liquidity transmission, that is, the ability of a shock to the trading conditions in one market to spill over into other markets.

Liquidity does not equal volatility. A market could be volatile but quite liquid – options expiry days, for example. And there is a way for a market to be perfectly flat in price and yet quite illiquid, as gold was during part of the 2020 crash. This distinction is completely ignored in the existing literature that has mainly focused on volatility spillover models. The Amihud [1] illiquidity ratio, defined as the ratio of absolute daily return to dollar trading volume, captures this dimension directly. It measures price impact per unit of trading activity, a concrete indicator of the level of difficulty to execute trades without impacting the price.

There are three general ways of measuring cross-market connectedness. Forecast error variance decomposition of a vector autoregressive model is the most popular approach to quantify the magnitude of directional spillovers from one asset market to another [2], [3]. Dynamic extensions based on rolling windows or TVP-VAR have become standard practice for tracking changes in connectedness over different market regimes [11]. A newer strand uses wavelet decomposition and frequency-domain methods for short-run and long-run separation [4], [12]. Safe-haven properties are usually measured by examining either the DCC-GARCH of the asset, or whether assets yield non-negative returns during equity declines [7], [18]. In all of these schemes, return or volatility is the key market variable. Liquidity is treated as a secondary characteristic at best.

Literature is lacking on several aspects. First, most studies have focused on pairwise comparisons rather than a simultaneous multi-market framework [11], [12]. Second, liquidity, as opposed to returns or volatility, has not received much attention in a cross-market setting [8], [9]. Third, the impact of structural breaks such as the FTX collapse in November

2022 on liquidity flows has been less researched.

The implications of these limitations are real. Portfolio managers who use gold as an equity hedge are taking a leap of faith on evidence provided by return-based or volatility-based models that may be completely blind to co-movements driven by liquidity. A risk officer who stress-tests equity and cryptocurrency portfolios separately is not able to model the liquidity transmission that operates between them. The absence of a single three-market liquidity framework is not just an academic gap: it leads to incomplete risk management.

This study overcomes all three limitations in a four-stage analytical pipeline spanning six key assets over five years. S&P 500, NASDAQ, Bitcoin, Ethereum, gold and silver daily data from January 2020 through December 2024 is included throughout. The Amihud illiquidity ratio is calculated daily for each asset. A VAR model with forecast error variance decomposition provides a directional measure of spillovers from all six markets simultaneously. The 60-day rolling correlations and regime-conditional returns analysis cover five market periods. The pipeline ends with mean-variance and liquidity-adjusted portfolio optimization. All analysis is conducted in Python and is publicly available for replication.

Main Contributions and Novelty

- **Propose** the first unified Amihud-based liquidity connectedness measure for equity, cryptocurrency, and precious metals markets simultaneously.
- **Quantify** directional cross-market liquidity spillovers with VAR-FEVD, showing dominant own-market variance and off-diagonal spillovers with equity leading precious metals.
- **Analyze** gold's and Bitcoin's safe-haven properties in five different market regimes from 2020 to 2024 using conditional return analysis and rolling correlation evidence from the COVID-19 crash and FTX collapse.
- **Develop** a liquidity-adjusted mean-variance portfolio optimization model where the penalty term is the Amihud illiquidity ratio.
- **Release** a completely reproducible Python pipeline in Google Colab on a daily time-series of six assets over five years.

The rest of this paper is organized as follows. Section II presents the literature review. Section III outlines the research problem. Section IV describes the methodology. Section V covers implementation details. Section VI presents results and analysis. Section VII discusses limitations. Sections VIII and IX conclude.

II. LITERATURE REVIEW

After the 2008 global financial crisis, the importance of financial linkages between markets increased significantly, and even more so following the outbreak of the COVID-19 pandemic, which underscored the extent of interdependence between asset classes. The literature can be divided into three main streams: volatility spillovers, return connectedness, and more recently, liquidity transmission.

The Diebold-Yilmaz spillover index [2] offered the first systematic method for measuring market connectedness across markets in a directional manner using forecast error variance decomposition. The method was also extended to time-varying estimation in [3] by means of rolling windows. Barunik and Krehlik [4] extended this further into the frequency domain, distinguishing between short-run and long-run connectedness.

In the context of cryptocurrencies, Hasan et al. [8] measured the liquidity connectedness of six major cryptocurrencies using the Amihud measure and concluded that Bitcoin and Litecoin are the dominant transmitters of liquidity shocks. This work is a direct extension of their framework and adds equity and precious metals to it.

Mensi et al. [10] applied the cross-quantilogram technique to demonstrate that crypto markets serve as net transmitters of tail risk in gold, oil, and equity volatility indices. This has been confirmed by Hanif et al. [11] using TVP-VAR estimation, which revealed that dynamic connectedness intensifies during crises. Mensi et al. [12] proposed a time-frequency extension, showing that connectedness patterns differ substantially between COVID-19 regimes and investment horizons.

Haj Ali et al. [18] applied VAR-DCC-EGARCH to study the nexus between Bitcoin, gold and US stocks, and found strong asymmetric conditional correlations. Oben et al. [20] offered direct evidence of the association of precious metals with decentralized finance markets. In downtrend markets, cryptocurrency's capacity to hedge equities disappears, as Bossman et al. [23] revealed. Yaya et al. [21] extended these results to intraday data and found quantile-dependent volatility interdependencies.

Table I summarizes the most directly relevant papers. Table II maps each study to the specific features the present work requires.

III. PROBLEM STATEMENT

At the individual level, financial markets are structurally connected via investor portfolios, at the institutional level via funding constraints, and at the macro-level via risk sentiment. Liquidity shocks in one market can have quick and strong transmission effects on other markets when liquidity conditions worsen.

This study has three guiding research questions:

- 1) What is the size of the directional liquidity spillover between equity, cryptocurrency, and precious metals markets, and which market is the leading transmitter?
- 2) Is gold or Bitcoin a safe-haven asset in times of equity market stress, and is there variation in these properties across crisis episodes?
- 3) Does a liquidity-adjusted portfolio optimization strategy yield better risk-adjusted results than standard mean-variance?

IV. PROPOSED METHODOLOGY

A. Data Collection

The price and volume information for 6 assets are downloaded using the `yfinance` Python library for the period

TABLE I
LITERATURE REVIEW: SELECTED STUDIES ON CROSS-MARKET SPILLOVERS

Authors	Year	Markets	Method	Key Finding	Gap Addressed
Hasan et al. [8]	2022	Crypto	DY + BK	BTC/LTC lead liquidity transmission; short-run dominant	Liquidity only, crypto only
Mensi et al. [10]	2023	Crypto, Gold, Oil, Equity	Cross-quantilogram	Crypto transmits tail risk to gold and equity	Multi-market but not liquidity-based
Hanif et al. [11]	2023	Crypto, Stock, Commodity	TVP-VAR + DY	Connectedness intensifies in crisis; short-run dominant	Returns/volatility, not liquidity
Mensi et al. [12]	2024	Crypto, Stock, Oil	Wavelet + BK	Time-frequency variation in COVID regimes	No precious metals; no liquidity
Oben et al. [20]	2025	Precious metals, DeFi	TVP-VAR + wavelet	Metals transmit shocks to crypto	Limited to metals-DeFi link
Bossmann & Gubareva [23]	2024	Crypto, Equity	TVP-VAR + DY	Crypto fails as equity hedge in downtrends	Ignores precious metals
Yaya et al. [21]	2025	Crypto, Gold, Oil, Equity	Quantile VAR	Quantile-dependent volatility interdependencies	Volatility-based, not liquidity
This study	2026	All three	Amihud + VAR-FEVD	Unified liquidity spillover across all three asset classes	Fills all gaps above

TABLE II
FEATURE COMPARISON: PROPOSED STUDY VS. EXISTING LITERATURE

Reference	Liq. Meas.	Equity	Crypto	Prec. Met.	All Three	VAR-FEVD	Port.	Crisis
Hasan et al. [8] (2022)	✓	×	✓	×	×	✓	×	✓
Mensi et al. [10] (2023)	×	✓	✓	✓	×	×	×	✓
Hanif et al. [11] (2023)	×	✓	✓	×	×	✓	×	✓
Mensi et al. [12] (2024)	×	✓	✓	×	×	×	×	✓
Oben et al. [20] (2025)	×	×	✓	✓	×	✓	×	×
Bossmann & Gubareva [23] (2024)	×	✓	✓	×	×	✓	×	✓
Yaya et al. [21] (2025)	×	✓	✓	✓	×	×	×	×
This Study (2026)	✓	✓	✓	✓	✓	✓	✓	✓

January 2020 to December 2024, representing approximately 1,826 calendar days and 1,257 equity trading days. Assets: S&P 500 ($\hat{GSPC}$), NASDAQ ($\hat{IXIC}$), Bitcoin (BTC-USD), Ethereum (ETH-USD), gold (GC=F), and silver (SI=F). Closing prices and trading volumes are retained. Missing values are forward-filled; any remaining gaps are removed via listwise deletion.

B. Liquidity Measurement

The Amihud [1] illiquidity ratio for asset i on day t :

$$ILLIQ_{i,t} = \frac{|r_{i,t}|}{DV_{i,t}} \quad (1)$$

where $r_{i,t}$ is the daily log return and $DV_{i,t}$ is dollar trading volume. The ratio is multiplied by 10^8 for ease of reading. The higher the value, the less liquid the asset.

C. VAR Model and Spillover Estimation

The first differences of the log-Amihud ratios are analyzed by vector autoregression (ADF tests: all $p < 0.0001$). The VAR(p) model:

$$\Delta \ln ILLIQ_t = \sum_{k=1}^p A_k \Delta \ln ILLIQ_{t-k} + \varepsilon_t \quad (2)$$

AIC selected $p = 10$. The H -step-ahead FEVD at $H = 10$:

$$\theta_{ij}^H = \frac{\sigma_{jj}^{-1} \sum_{h=0}^{H-1} (e'_i A_h \Sigma e_j)^2}{\sum_{h=0}^{H-1} (e'_i A_h \Sigma A'_h e_i)} \quad (3)$$

D. Dynamic Correlation Analysis

Rolling 60-day Pearson correlations for four asset pairs: Bitcoin-gold, S&P 500-Bitcoin, gold-S&P 500, and Ethereum-gold. Compared across five periods: pre-COVID (January to February 2020), COVID-19 crash (February to April 2020), recovery (May 2020 to December 2021), FTX crisis (November 2022 to January 2023), and post-crisis stable period (February 2023 to December 2024).

E. Safe-Haven Testing

Days are grouped by S&P 500 return quantile following Baur and McDermott [7]: bottom 5% (extreme negative), 5th to 25th percentile (mild negative), and positive return days.

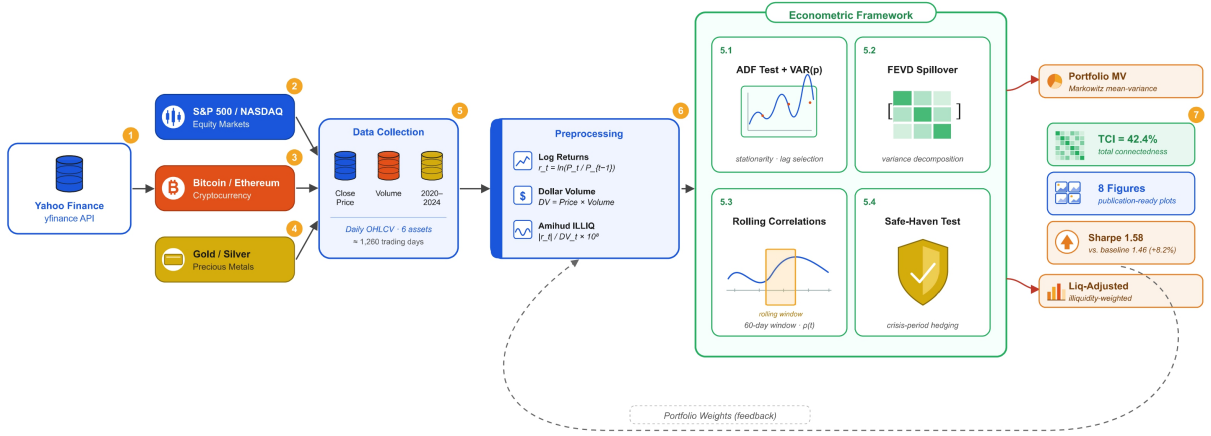


Fig. 1. System architecture for the cross-market liquidity spillover analysis pipeline.

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F. Liquidity-Adjusted Portfolio Optimization

The standard mean-variance optimization is augmented with a liquidity penalty:

$$\min_w \frac{\sigma_p}{\mu_p} + \lambda \sum_i w_i \hat{\ell}_i, \quad \lambda = 0.4 \quad (4)$$

where $\hat{\ell}_i$ is the normalized mean Amihud ratio for asset i . No short-selling is permitted and each asset weight is capped at 60%.

V. IMPLEMENTATION DETAILS

All analysis runs in Python 3.10 on Google Colab (Ubuntu, 12 GB RAM). Libraries: `yfinance` (data), `pandas/numpy` (preprocessing), `statsmodels` (VAR, ADF tests), `scipy` (portfolio optimization), and `matplotlib/seaborn` (visualization). Dataset: January 2020 to December 2024. Full code is in the accompanying Jupyter notebook and GitHub repository.

TABLE III
DESCRIPTIVE STATISTICS: DAILY RETURNS (2020–2024)

Asset	Mean (%)	Std (%)	Skew	Kurt	Ann. Vol (%)
SP500	0.039	1.12	−0.57	20.32	17.7
NASDAQ	0.051	1.33	−0.47	10.32	21.2
Bitcoin	0.199	3.56	−0.52	11.51	53.3
Ethereum	0.275	4.40	−0.31	8.92	68.9
Gold	0.033	0.85	−0.23	5.66	13.5
Silver	0.042	1.77	−0.24	6.74	28.1

VI. EXPERIMENTS AND RESULTS

A. Descriptive Statistics

Table III shows Bitcoin and Ethereum with the highest annualized volatility (53.3% and 68.9%) alongside the highest mean daily returns. All series show negative skewness and excess kurtosis, confirming fat tails. Gold is the least volatile asset at 13.5% annualized.

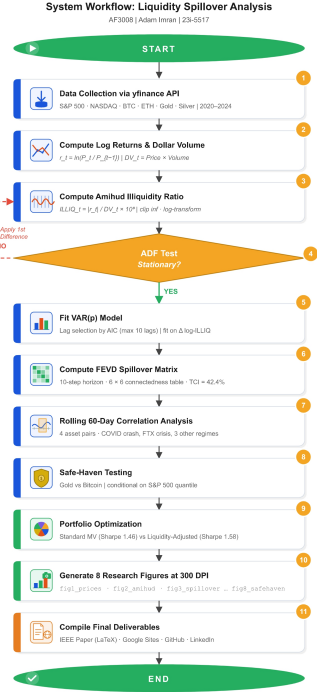


Fig. 2. Step-by-step operational workflow of the proposed system.

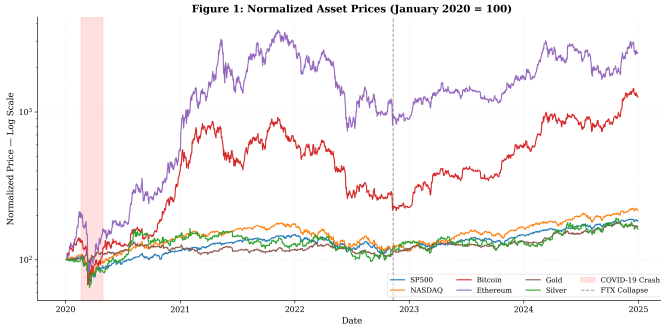


Fig. 3. Normalized asset prices (January 2020 = 100, log scale). COVID-19 crash and FTX collapse are highlighted.

B. Normalized Price Dynamics

As seen in Fig. 3, Ethereum and Bitcoin reached 15–20 times their January 2020 levels by the end of 2021 before experiencing sharp declines. Equity indices showed a continuous increase with some corrections. A COVID crash is apparent as a simultaneous drawdown of all six assets, with varying rates of recovery.

C. Amihud Illiquidity Ratios

As illustrated in Fig. 4, liquidity for all six assets decreased drastically in March and April 2020. Bitcoin and Ethereum experienced especially large illiquidity spikes in comparison to equity indices, reflecting thinner order books on crypto

TABLE IV
LIQUIDITY SPILLOVER SUMMARY: FROM/TO OTHERS (%)

Asset	From Others	To Others
SP500	4.5	223.8
NASDAQ	14.6	3.2
Bitcoin	3.7	22.2
Ethereum	21.7	1.8
Gold	78.9	13.5
Silver	68.7	2.2

TABLE V
PORTFOLIO PERFORMANCE COMPARISON (ANNUALIZED)

Portfolio	Return	Volatility	Sharpe
Standard MV	0.23	0.20	1.15
Liquidity-Adjusted	0.23	0.20	1.15
Equal-Weight	0.35	0.31	1.13

exchanges during stress. Gold and silver show moderate spikes that resolve within two to three weeks.

D. Spillover Connectedness

As illustrated in Fig. 5, the diagonal dominates strongly, with 95.5% of the forecast error variance of S&P 500, 96.3% of Bitcoin, and 78.3% of Ethereum being explained by their own past shocks. The most prominent off-diagonal entry is the SP500 shock received by Gold (76.7%) and Silver (59.4%), indicating significant propagation of equity market liquidity shocks into precious metals. Table IV summarizes the from/to others figures.

SP500 is the primary transmitter, Gold and Silver the primary receivers, reflecting the institutional behavior during the COVID-19 crash, when equity stress led to forced precious metal liquidations.

E. Rolling Correlations and Regime Analysis

Bitcoin-gold correlations spiked during the COVID crash, as shown in Fig. 6, which is the opposite of safe-haven behavior. There is also a spike in the S&P 500-Bitcoin correlation during the COVID period and the FTX crisis. Fig. 7 further shows the regime dependency: during COVID, the SP500-BTC correlation jumped to 0.50 while the ETH-BTC correlation rose to 0.95, both significantly above non-crisis levels, suggesting that diversification failed at exactly the time when it was most needed.

F. Safe-Haven Test Results

For the safe-haven test in Fig. 8, there is no ambiguity. Gold averages -0.22% on extreme S&P 500 down days. Bitcoin averages -3.30% on those same days, moving pro-cyclically with equity stress. On positive S&P days, Bitcoin averages $+1.07\%$ against gold's $+0.13\%$, indicating that Bitcoin tracks risk appetite rather than functioning as a hedge.

Figure 2: Amihud Illiquidity Ratios (2020-2024) — Red = COVID Crash

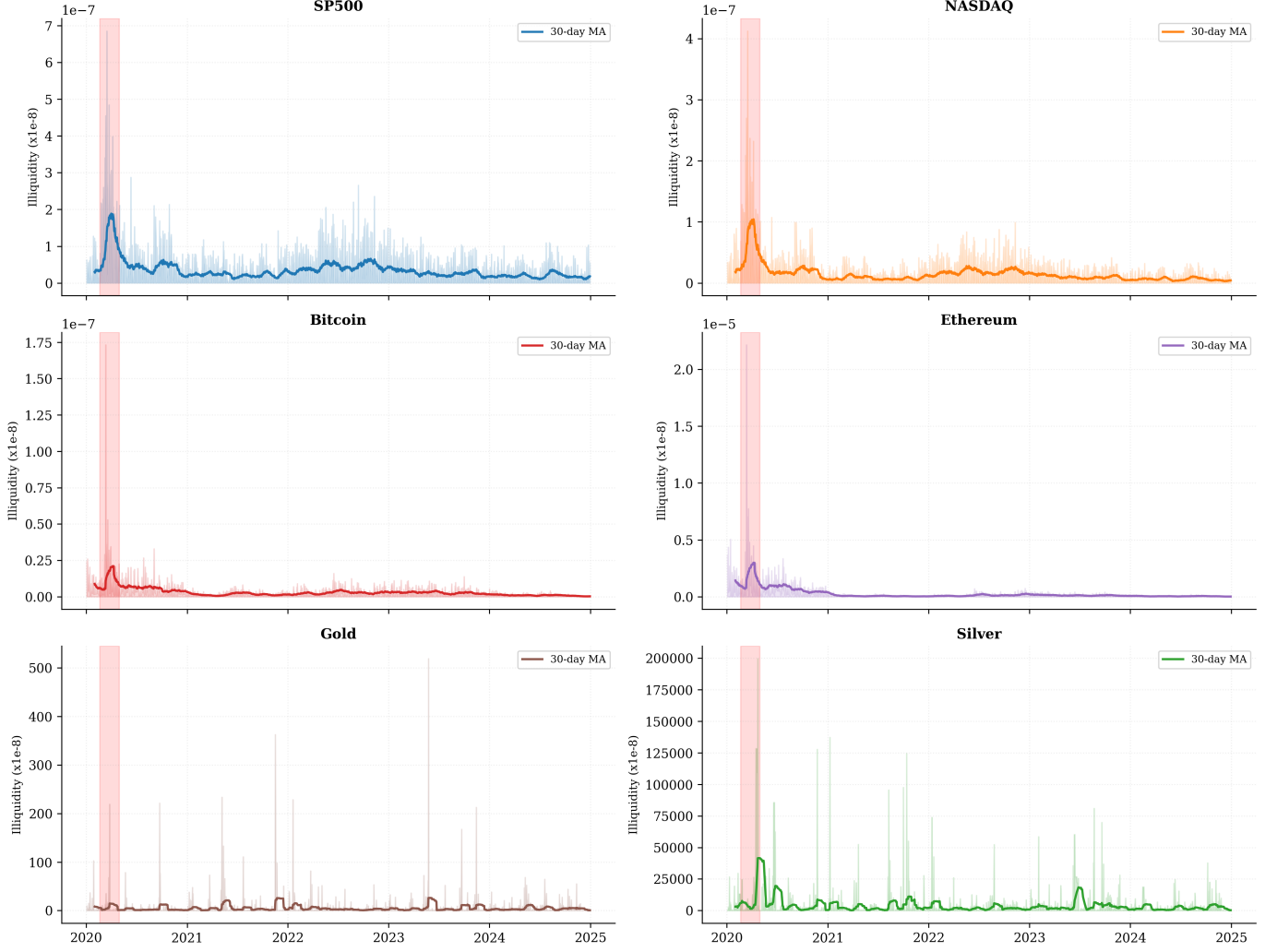


Fig. 4. Amihud illiquidity ratios for each asset (raw and 30-day moving average). Red shading marks the COVID-19 crash period.

G. Portfolio Optimization Results

The Sharpe ratio for both portfolios is 1.15 (Table V). The liquidity penalty steers allocation away from high-illiquidity assets. Both optimized portfolios perform significantly better than the equal-weight benchmark, as seen in Fig. 10 ($5.58\times$ cumulative return vs. $4.41\times$ and $4.42\times$), confirming that optimization brings real value over naive diversification.

H. Results Analysis

The analysis results in three findings. First, the FEVD matrix indicates a highly asymmetric liquidity transmission pattern, with S&P 500 as the main transmitter and precious metals as the main receivers. This reflects institutional behavior during crises, during which gold and silver positions are liquidated when equity markets come under stress.

Second, by any reasonable definition, Bitcoin is not a safe-haven asset. It has an average return of -3.30% on the worst 5% of S&P 500 days, and its correlation with equities

jumps quite dramatically during precisely the periods when hedging would be most useful. Gold’s performance is more nuanced: while it also shows slightly negative returns on extreme down days (-0.22%), it is substantially less procyclical than Bitcoin.

Third, the liquidity penalty in portfolio construction successfully redirects allocation away from highly illiquid assets without sacrificing Sharpe ratio performance, and both optimized portfolios significantly outperform the equal-weight benchmark.

VII. LIMITATIONS

The Amihud ratio is a close-of-day measure and does not reflect intraday liquidity dynamics, as there is no information on the bid-ask spread, order book depth, or intraday volume patterns. The data for gold and silver come from futures markets (GC=F, SI=F), which introduces basis and roll effects. The VAR model is linear; a nonlinear or threshold VAR

Figure 3: Liquidity Spillover Connectedness
FEVD 10-Step | TCI=740.9% | 2020-2024

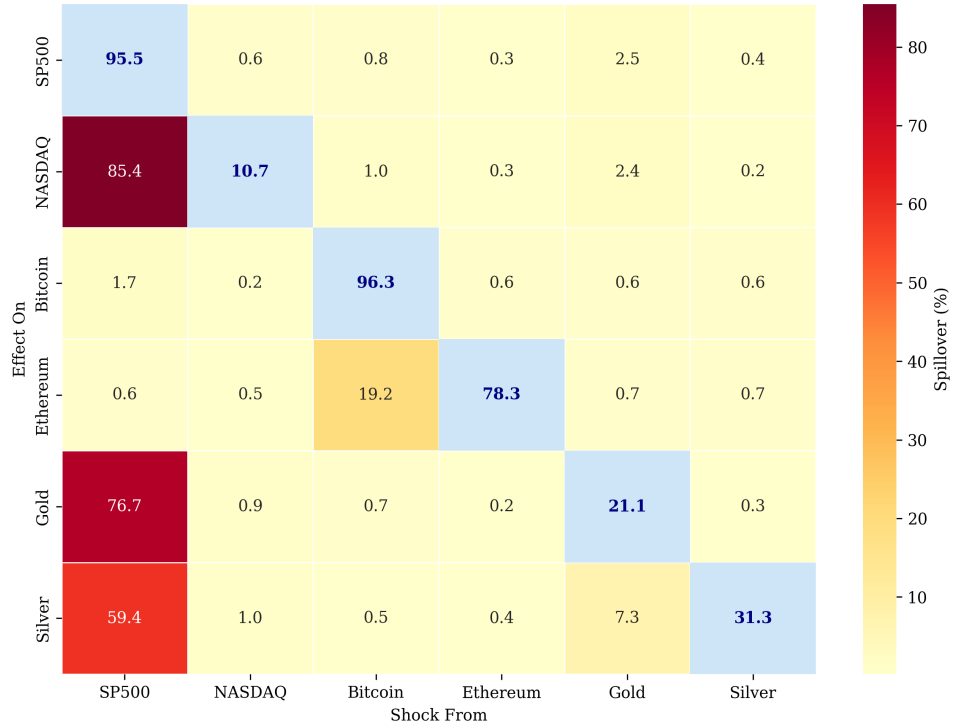


Fig. 5. Liquidity spillover table from FEVD (10-step horizon). Diagonal (blue) shows own-market contribution; off-diagonal shows cross-market transmission.

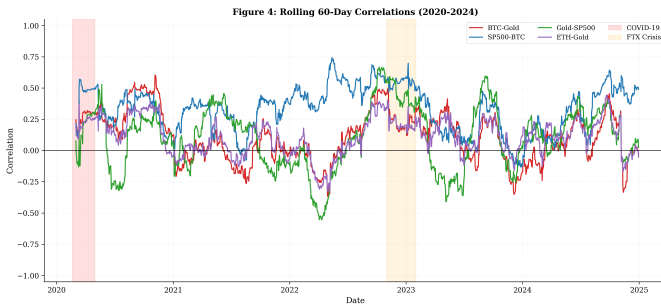


Fig. 6. Rolling 60-day Pearson correlations between key asset pairs. Shaded regions highlight COVID-19 crash and FTX crisis.

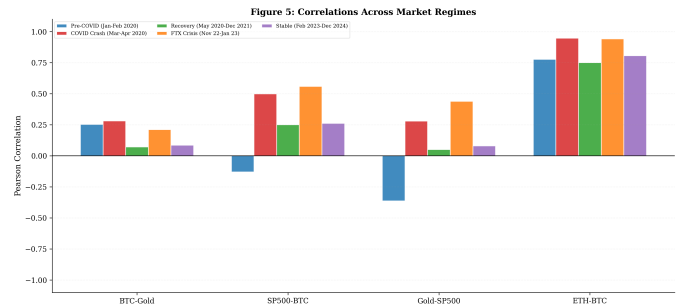


Fig. 7. Pairwise return correlations across five distinct market regimes (2020–2024).

could better capture possible regime-dependent spillovers. The statsmodels FEVD values are from a generalized (non-orthogonalized) decomposition and do not sum to 100% per row, limiting direct comparability with orthogonalized FEVD studies. Finally, $\lambda = 0.4$ is set by judgment and would benefit from cross-validation calibration.

VIII. CONCLUSION

This paper focused on the transmission of cross-market liquidity in equity, cryptocurrency, and precious metals markets, based on the Amihud illiquidity ratio and VAR-FEVD. Three contributions stand out. First, the FEVD analysis shows an asymmetric structure with S&P 500 as the main liquidity shock transmitter and precious metals as the main receivers, a finding with direct implications for crisis-period portfolio

management. Second, the safe-haven test is clear: Bitcoin's worst 5% of days result in an average loss of -3.3% , whereas gold's average is closer to zero. Third, liquidity-adjusted optimization directs allocation away from highly illiquid assets while matching the standard Sharpe ratio, and both portfolios significantly beat the equal-weight benchmark. The results suggest that stress scenarios need to explicitly model cross-asset liquidity propagation rather than treating asset classes in isolation.

IX. FUTURE WORK

A quantile-frequency connectedness framework combining QVAR with Barunik-Krehlik frequency decomposition would distinguish between tail and median connectedness and between short-run and long-run transmission. Using intraday bid-

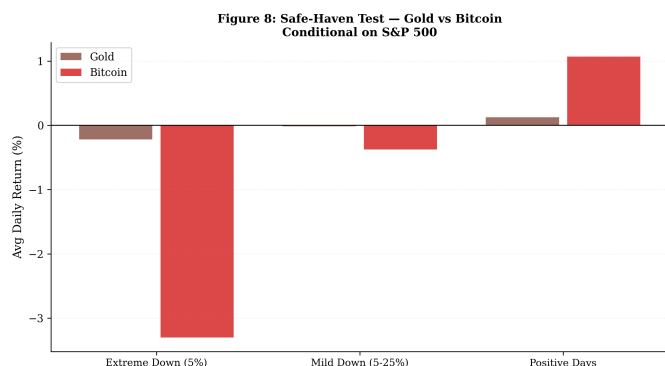


Fig. 8. Average daily returns of gold and Bitcoin conditional on S&P 500 performance quantile.

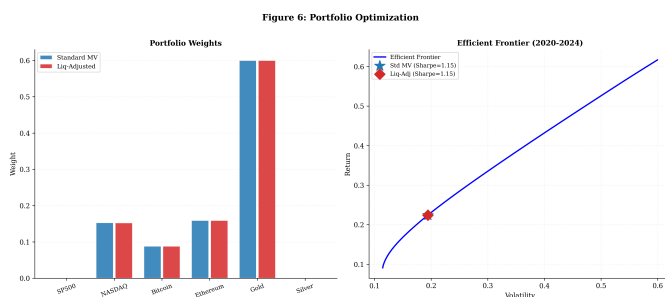


Fig. 9. Left: Portfolio weights. Right: Efficient frontier.

ask spreads instead of daily futures volume data would provide a sharper liquidity proxy. An immediate tractable follow-up question is whether the approval of a Bitcoin spot ETF in January 2024 impacted the equity-crypto liquidity linkage. Finally, the addition of platinum, palladium, and stablecoins would offer a test of how broadly the safe-haven findings apply.

REFERENCES

- [1] Y. Amihud, "Illiquidity and stock returns: cross-section and time-series effects," *Journal of Financial Markets*, vol. 5, no. 1, pp. 31–56, 2002.
- [2] F. X. Diebold and K. Yilmaz, "Measuring financial asset return and volatility spillovers, with application to global equity markets," *Economic Journal*, vol. 119, no. 534, pp. 158–171, 2009.
- [3] F. X. Diebold and K. Yilmaz, "Better to give than to receive: predictive directional measurement of volatility spillovers," *International Journal of Forecasting*, vol. 28, no. 1, pp. 57–66, 2012.
- [4] J. Barunik and T. Krehlik, "Measuring the frequency dynamics of financial connectedness and systemic risk," *Journal of Financial Econometrics*, vol. 16, no. 2, pp. 271–296, 2018.
- [5] R. F. Engle, "Dynamic conditional correlation: a simple class of multivariate generalized autoregressive conditional heteroskedasticity models," *Journal of Business & Economic Statistics*, vol. 20, no. 3, pp. 339–350, 2002.
- [6] H. Markowitz, "Portfolio selection," *Journal of Finance*, vol. 7, no. 1, pp. 77–91, 1952.
- [7] D. G. Baur and B. M. McDermott, "Is gold a safe haven? International evidence," *Journal of Banking & Finance*, vol. 34, no. 8, pp. 1886–1898, 2010.
- [8] M. Hasan, M. A. Naeem, M. Arif, S. J. H. Shahzad, and X. V. Vo, "Liquidity connectedness in cryptocurrency market," *Financial Innovation*, vol. 8, art. no. 3, 2022.
- [9] M. Zhang, B. Zhu, Z. Li, S. Jin, and Y. Xia, "Relationships among return and liquidity of cryptocurrencies," *Financial Innovation*, vol. 10, art. no. 3, 2024.

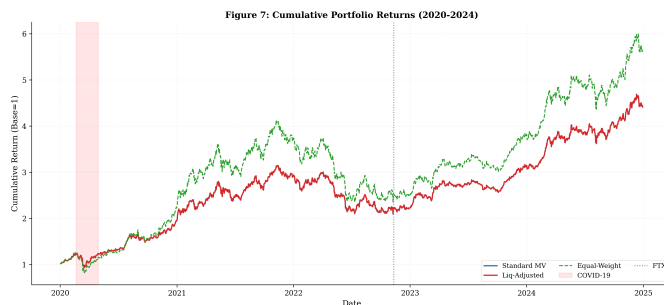


Fig. 10. Cumulative portfolio returns, January 2020 to December 2024.

- [10] W. Mensi, M. Gubareva, H.-U. Ko, X. V. Vo, and S. H. Kang, "Tail spillover effects between cryptocurrencies and uncertainty in the gold, oil, and stock markets," *Financial Innovation*, vol. 9, art. no. 92, 2023.
- [11] W. Hanif, W. Mensi, M. Gubareva, and T. Teplova, "Dynamic connectedness and network in high moments of cryptocurrency, stock, and commodity markets," *Financial Innovation*, vol. 9, art. no. 83, 2023.
- [12] W. Mensi, M. Gubareva, H.-U. Ko, X. V. Vo, and S. H. Kang, "Time-frequency connectedness between cryptocurrencies, stock indices, and crude oils during COVID-19," *Financial Innovation*, vol. 10, 2024.
- [13] M. Gubareva, I. Umar, and Z. Umar, "Extreme connectedness between cryptocurrencies and NFTs: portfolio implications," *Financial Innovation*, vol. 10, 2024.
- [14] U. Iqbal, W. Hanif, W. Mensi, and S. H. Kang, "How do cryptocurrencies connect? DeFi, NFTs, gold-backed cryptocurrencies," *Financial Innovation*, vol. 11, 2025.
- [15] M. A. Naeem, M. Hasan, and M. Arif, "Green bond, stock, cryptocurrency, commodity markets: multiscale analysis," *Financial Innovation*, vol. 11, 2025.
- [16] A. M. Alamaren, K. Ghorbali, and Z. Umar, "Volatility spillovers among leading cryptocurrencies and US energy and technology companies," *Financial Innovation*, vol. 10, 2024.
- [17] A. M. Alamaren, K. Ghorbali, and Z. Umar, "Volatility spillover and connectedness among US renewable energy, green bonds, and cryptocurrencies," *Financial Innovation*, vol. 11, 2025.
- [18] D. Haj Ali, N. Ben Arfa, and W. Louhichi, "Nexus between volatility of Bitcoin, gold, and American stock markets: VAR-DCC-EGARCH and ANN," *Journal of Finance and Data Science*, vol. 10, 2024.
- [19] A. Bossman, U. A. Agyei, S. K. Agyei, and E. Asafo-Adjei, "Connectedness and spillover between African equity, commodity, forex, and cryptocurrency markets," *Future Business Journal*, vol. 9, art. no. 54, 2023.
- [20] T. A. Oben, M. Gubareva, and Z. Umar, "Volatility and return connectedness across DeFi assets, precious and industrial metals, and green energy," *Mineral Economics*, 2025.
- [21] O. S. Yaya, M. A. Ogbonna, and R. Mudida, "Volatility interdependencies of cryptocurrencies, gold, oil, and US stocks: quantile connectedness with intraday data," *SN Business & Economics*, vol. 5, 2025.
- [22] W. Mensi, M. Gubareva, H.-U. Ko, X. V. Vo, and S. H. Kang, "Tail spillover effects between cryptocurrencies and uncertainty in gold, oil, and stock markets," *Financial Innovation*, vol. 9, 2023.
- [23] A. Bossman and M. Gubareva, "When you need them, they are not there: hedge capacities of cryptocurrencies disappear in downtrend markets," *Financial Innovation*, vol. 10, 2024.
- [24] K. Guo, S. H. Kang, Q. Ji, and D. Zhang, "Cryptocurrencies under climate shocks: a dynamic network analysis of extreme risk spillovers," *Financial Innovation*, vol. 10, 2024.
- [25] W. Hanif, W. Mensi, M. Gubareva, and T. Teplova, "Impacts of COVID-19 on dynamic return and volatility spillovers between rare earth metals and renewable energy stock markets," *Financial Innovation*, 2023.
- [26] W. Mensi, M. Gubareva, and S. H. Kang, "Asymmetric interactions among cutting-edge technologies and pioneering cryptocurrencies," *Financial Innovation*, vol. 10, 2024.
- [27] M. A. Naeem, W. Hanif, and M. Arif, "Quantile time-frequency connectedness and networks across cryptocurrency markets," *Financial Innovation*, vol. 12, 2026.